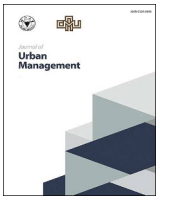
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Research Article

Urban heat in Mumbai: From scenario analysis to the mitigation of nighttime thermal footprint



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ABSTRACT

Rapid urbanisation has increased urban land surface temperatures, emerging as a critical environmental challenge in megacities such as Mumbai. Urban nighttime land surface temperature (LST) is associated with the Urban Heat Island (UHI) effect, leading to urban thermal stress, health hazards, and energy usage. Despite extensive research on urban LST, there remains a critical lack of spatially adaptive modelling frameworks capable of effectively capturing fine-scale local variations in environmental and anthropogenic drivers of LST, particularly in complex megacities such as Mumbai. This study bridges this gap by utilising satellite-derived datasets and advanced geospatial modelling techniques, including Ordinary Least Squares (OLS), Geographically Weighted Regression (GWR), and Random Forest (RF), to map the intensity, factors, and temporal dynamics of nighttime LST and UHI in the Mumbai Metropolitan Region (MMR) from 2000 to 2025. Temporal analysis reveals a steady increase in nighttime LST in built-up zones, with LST differentials reaching up to +5.07 °C in the southern and central MMR. Nighttime light emerges as the dominant factor (R^2 = 0.826), while NDVI (vegetation) and wind speed show strong negative correlations (r = −0.243 and r = −0.052). The GWR model outperforms both OLS and RF, achieving an R^2 of 0.9808 and an Adjusted R^2 of 0.9539, offering a more effective tool for capturing local spatial variations in LST. The study suggests innovative mitigation strategies, such as urban greening and reflective surfaces, alongside smart lighting systems, for alleviating UHI impacts. This research provides a valuable framework for addressing the UHI effect, not only in Mumbai but also in other rapidly urbanising cities across the Global South.

1. Introduction

The Urban Heat Island (UHI) effect is one of the most vital expressions of climate variation in urban environments. The global mean temperature is rapidly increasing by approximately 0.06 °C per decade, primarily driven by increased greenhouse gas emissions (GHGs), specifically carbon dioxide, and strengthening anthropogenic activities (Lindsey, Dahlman, LuannDahlman, & Blunden, 2024). Moreover, anthropogenic activities are transforming the natural landscape into a concrete one, influencing thermal stress in the urban environment compared with the surrounding rural environment (Jain et al., 2020, 2022). The UHI concept is generally recognised as a key indicator of urban climate change, indicating the complexity of the urban climatic variety (Abulibdeh, 2021; Hou

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et al., 2023). Inside the urban spaces, unique thermal and microclimatic conditions occur, driven by environmental thermal dynamics (Jain et al., 2020). Rising temperatures exacerbate heat stress vulnerability, impacting urban settlements, quality of life, and the environmental conditions affecting human health in cities (Santamouris, 2020), as well as the economy of the country (Wei & Sobrino, 2024).

Rising temperatures in urban spaces also worsen the occurrence of dangerous air pollutants, while increasing energy consumption, compromising air quality, elevating health risks, and expanding heat-related mortality (Santamouris, 2020; Santamouris et al., 2022). Moreover, rising temperatures contribute to higher energy demand for cooling, leading to additional greenhouse gas emissions (Wei & Sobrino, 2024). The UHI Intensity (UHII) depends on a combination of factors, such as meteorological and climatic conditions, demographic trends, urban area size and density, morphology, urban design, availability of water bodies and land reclamation, land use and land cover (LULC) characteristics, green space proportions, anthropogenic heat generation, while building material types determine the thermal equilibrium and UHI effect (Abulibdeh, 2021). The UHI effect varies worldwide due to the particular locations and features of different cities; each of the listed elements is distinct for different parts of the world according to the findings from 20-year regional convection-permitting model simulations that encompass the UHI effect for future climates in the megacities of Paris and Shanghai (Mushtaha et al., 2021; Steensen et al., 2022). The UHI phenomenon is exacerbated by the rapid rate of urbanisation, as well as the city's coastal location, fostering a complex interaction between local climate, urban form, and human activity. Therefore, the UHI is a dynamic urban concept; its effects vary due to different features among various cities. Mumbai demonstrates unique challenges that require targeted and innovative UHI mitigation strategies and planning, shaped by its tropical and sub-tropical climate, dense urban settlements, and rapid land development. Most of the recent inland metro-cities studies have focused on Mumbai's UHI due to its maritime climate, sea-breeze circulation, and anthropogenic changes, which strongly influence and shape the characteristic spatial heat patterns throughout the Mumbai Metropolitan Region (MMR). However, during the summer, the demand for cooling energy consumption rises in the city, increasing heat-related illness and reported mortality due to heatstroke. Rising temperatures also directly and indirectly affect human health and air quality (Santamouris, 2020; Santamouris et al., 2022).

The evidence shows that the surface temperature in the MMR can be 4 to 5 °C higher than in nearby rural settings during the summer (Jain et al., 2020, 2022). Mumbai's urban form of densely packed buildings, small streets, and little natural cover makes it quite vulnerable to the UHI effect, given its tropical environment (Nandineni et al., 2024; Patel & Shah, 2023). Furthermore, the worsening urban heat impact is attributed to the rapid transformation of land use from open spaces to developed regions, increasing air pollution, and reducing the city's natural cooling capacity (Ansari et al., 2025; Pande et al., 2024). Urban geometry, LULC, and climate variability are the main anthropogenic and environmental drivers of UHII in Mumbai. The increasing percentage of impermeable surfaces, such as asphalt and concrete, which absorb and store heat, may help to explain the rise in metropolitan temperatures. The restricted availability of green areas and flora further lowers the natural heat dissipation mechanisms (García, 2022). On the other hand, due to the cooling effect of vegetation through evapotranspiration (Rahmani & Sharifi, 2023), locations with higher green cover, such as parks and tree-lined avenues, tend to have lower UHII (Pramanik et al., 2025; Tun et al., 2025). Particularly in Mumbai slums, the high population density exacerbates the UHI impact, as more human activity generates greater anthropogenic heat (Patel & Shah, 2023). Children, women, and the elderly are more vulnerable, and for individuals with pre-existing medical conditions, exposure to UHI effects has serious health consequences. Cities with higher UHI effects are seeing more and more heat-related diseases, including heatstroke and dehydration (Bohra et al., 2020; Chakraborty & Lee, 2019). Higher temperatures contribute to the production of ground-level ozone, a hazardous pollutant that exacerbates respiratory ailments and cardiovascular disorders, thereby worsening the air pollution situation in the city (Chakrabortty et al., 2025). Economically, the surge in energy demand for cooling purposes such as refrigeration and air conditioning strains the power supply of cities, hence equally raising consumer and commercial expenses (Siddiqui et al., 2021). Expanding urban vegetation, including parks, green roofs, and urban forests, is among the most effective strategies for reducing UHII (Venter, Aunan, et al., 2021; Venter, Chakraborty, & Lee, 2021). Moreover, the use of green roofing helps buildings to remain cool by minimising heat absorption, providing an instant process for reducing the temperature (Golany, 1996; Brown et al., 2021). Urban planning policies must consider the conservation aspects and focus on increasing the natural space within the city, such as with wetlands, public parks, and the implementation of nature-based solutions in infrastructural development to reduce nature-infused thermal regulation and environmental balance (Tun et al., 2025; Chakraborty et al., 2024).

Recent developments in GIS and remote sensing have significantly improved the analysis and assessment of UHI models and their impact on cities like Mumbai. The monitoring of LST and LULC alterations by satellite data facilitates the accurate identification of UHI hotspots and informs urban development strategies (Jain et al., 2020; Santamouris et al., 2022; Shafizadeh Moghadam and Helbich, 2013). Most previous research has focused on LST, UHI, and a limited set of biophysical indicators, such as the Normalised Difference Built-up Index (NDBI), Normalised Difference Vegetation Index (NDVI), Normalised Difference Water Index (NDWI), and land use/-land cover (LULC). These studies mostly focus on correlation-based assessment and analysis between LST and land surface features. However, they pay less attention to different demographic and atmospheric factors such as population density, aerosol optical depth (AOD), surface albedo, wind speed, slope, and nighttime light intensity, despite their direct and indirect impact on how heat builds up and dissipates in cities. Therefore, this study introduces a new framework to elucidate the LST and UHI dynamics by using both conventional biophysical indicators and an expanded array of urban-atmospheric and socio-environmental variables. Furthermore, the LST is considered an outcome variable, whereas the exposure variables are socio-demographic, atmospheric, topographic, and anthropogenic factors. This study develops three statistical techniques, namely Ordinary Least Squares (OLS), Geographically Weighted Regression (GWR), and Random Forest (RF), enabling the capture of both global and local interactions. The OLS model describes the overall effect of each exposure variable on LST at the global level, while GWR examines spatial variations in the influence of exposure variables on LST on a more localised scale. Additionally, the RF model examines the nonlinear interactive relationship among all exposure variables and identifies those most influentially impacting the intensity of LST. Jointly applying the analysis

techniques identified through global and local regression analyses, GWR, and machine learning provides a comprehensive framework for studying both the magnitude and spatial distribution of the factors influencing LST. The study proposes the mechanisms by which urbanisation and climate change affect the variations in heat in tropical megacities, such as Mumbai, based on long-term time series and the use of hybrid spatial models. By analysing the trends of Mumbai UHI in comparison with global and regional contexts, this research represents a significant breakthrough in comprehending the issue and offers an escape route to sustainable urban planning, green infrastructure construction, and adaptation strategies at the community level.

2. Study area

Mumbai is one of the largest coastal urban centres in western India, extending between 18°33' and 19°31' North and 72°45' and 72°28' East (Fig. 1). Globally, Mumbai is the sixth largest city based on the total population, accounting for approximately 22.8 million people living within an area of 458.28 sq km (Pramanik et al., 2021). This urban area is characterised by high population density, with 5361 individuals per sq km, and the highest observed in Greater Mumbai and Bhiwandi, followed by Ulhasnagar (Census of India, 2011). This city comes under the tropical wet and dry climate, where most of the rainfall is received during the monsoon season (June to September), the annual average rainfall is 2200 mm, and the average temperature varies from 25 °C to 32 °C (Pramanik, 2017). Mumbai is an important part of India's economy as the nation's commercial and financial hub. The ecology, land use patterns, and infrastructure of the city have been profoundly damaged by uncontrolled expansion, high population density, industrial growth, and rapid urbanisation (Kavathekar et al., 2024; Mangaraj et al., 2024; Tripathy et al., 2024). The spread of slums, traffic jams, pollution of the air and water, and growing susceptibility to climate-related risks, including urban flooding and sea level rise, are some of the major urban issues. Studies by Rahaman, Jahangir, Haque, Chen, and Kumar (2021a) and Shahfahad et al. (2022) emphasise the strong linkage between rising LST across MMR with an increase in built-up areas and declining vegetation cover. While urban growth and

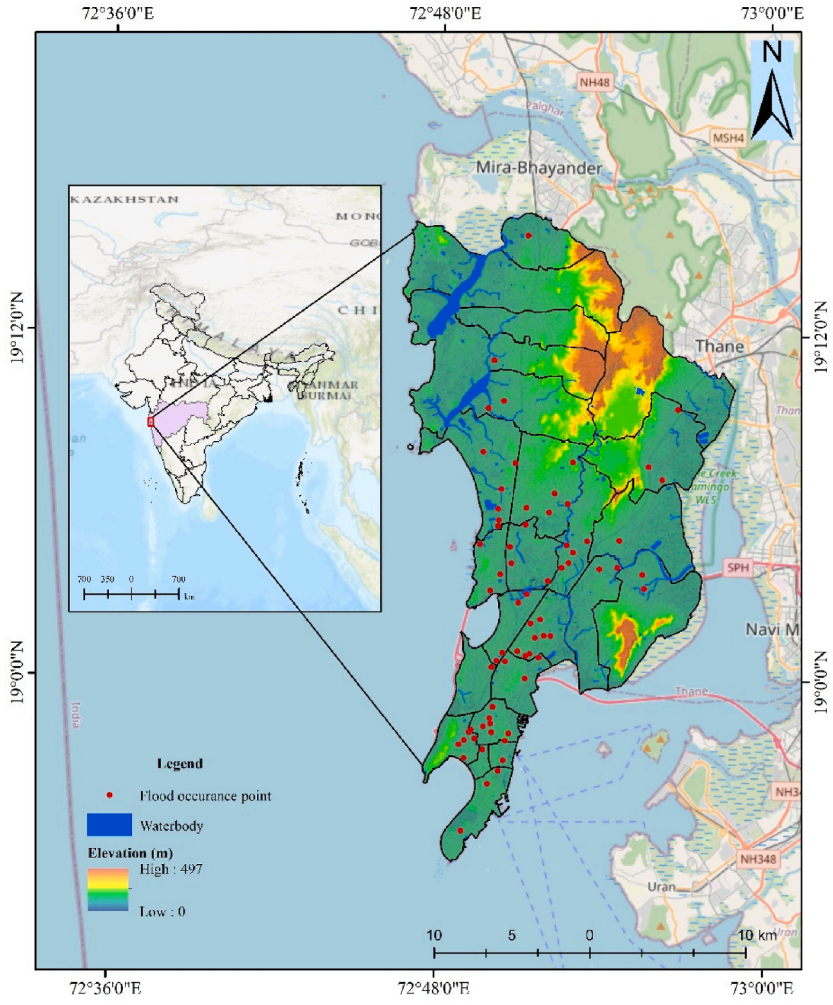


Fig. 1. Map showing the boundary and a satellite image of the study area.

heat have been addressed through spatial regression studies, an inclusive assessment integrating both natural and anthropogenic variables remains unexplored. Therefore, the present study proposes the use of MODIS nighttime LST data to quantify the spatial dynamics of urban heat intensity at night in MMR and examine the role of various socioeconomic and environmental parameters. The methodological framework presented in this paper can be adapted for application to other megacities facing similar heat-related issues.

3. Methods

The methodology of this study examined the association between several environmental and anthropogenic factors with urban nighttime heat in the MMR by integrating remote sensing and geospatial analysis techniques (Fig. 2). Initially, all the raw data extracted within the Mumbai study area boundary were also used to fix the projection system WGS 1984 UTM Zone 43N, along with data downscaling and resampling. The pixel values were then extracted from the resampled data using the zonal statistics tool in ArcGIS software. The strength and significance of the correlations between nighttime LST and the chosen variables were measured using RF, OLS, GWR, and Pearson correlation analysis.

3.1. Dataset

The current study employed a wide range of remote sensing and geospatial datasets to examine the urban environmental dynamics (Table 1). Nighttime LST and surface albedo data for the year 2024 were obtained from the MODIS. Specifically, LST data were collected from the MOD11A2 product at a spatial resolution of 1 km, and albedo values were sourced from the MCD43A3 product, which offers 500-m resolution. UHII was assessed using both MOD11A2 and MYD11A2 LST products by comparing urban and adjacent non-urban areas. The LULC, NDVI, NDBI, and NDWI data were extracted from the Landsat 8 Operational Land Imager (OLI), with 30-m spatial resolution, accessible through the USGS Earth Explorer portal. Nighttime light data, indicating anthropogenic activity levels, were obtained from the VIIRS Day/Night Band (DNB) monthly composites, provided by the National Oceanic and Atmospheric Administration (NOAA), with an approximate resolution of 500 m. Wind speed data were retrieved from the ERA5 dataset maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF), which carries out reanalysis globally with approximately 0.25° spatial resolution. AOD, used as a single scalar parameter to identify pollution in the atmosphere, was gathered from MODIS MOD04_L2 data with a 10 km resolution. Slope data were constructed using the Shuttle Radar Topography Mission Digital Elevation Model (SRTM DEM) available at 30 m spatial resolution. In addition, gridded population density data were collected for 2020 from WorldPop at a fine resolution of 100 m.

3.2. Data preprocessing

All the satellite-based datasets were prepared using the Google Earth Engine (GEE) platform, including MODIS-derived nighttime LST, surface albedo, and AOD, as well as Landsat 8 imagery for NDVI, NDBI, NDWI, and LULC classification. Nighttime light data, wind speed, slope from SRTM DEM, and population density from WorldPop were also acquired through GEE. After preprocessing in GEE, all raster layers were imported into the ArcGIS software for further spatial analysis. All the raw images were reprojected to a common coordinate reference system, specifically WGS 1984 UTM Zone 43N, to ensure spatial consistency across all variables. To standardise the resolution, all datasets were resampled to a 1000-m resolution using the nearest neighbour method, based on the MODIS LST raster. Following resampling and downscaling, raster-to-point conversion was applied to the LST layer to generate spatial point features. Using the "Extract Multi Values to Points" tool in ArcGIS, values from all pixels were extracted for each point. This generated a comprehensive geospatial dataset with LST values and corresponding influencing factors at each point location. Statistical analyses were then conducted using both spatial and machine learning techniques. OLS and GWR were applied within ArcGIS to explore spatial

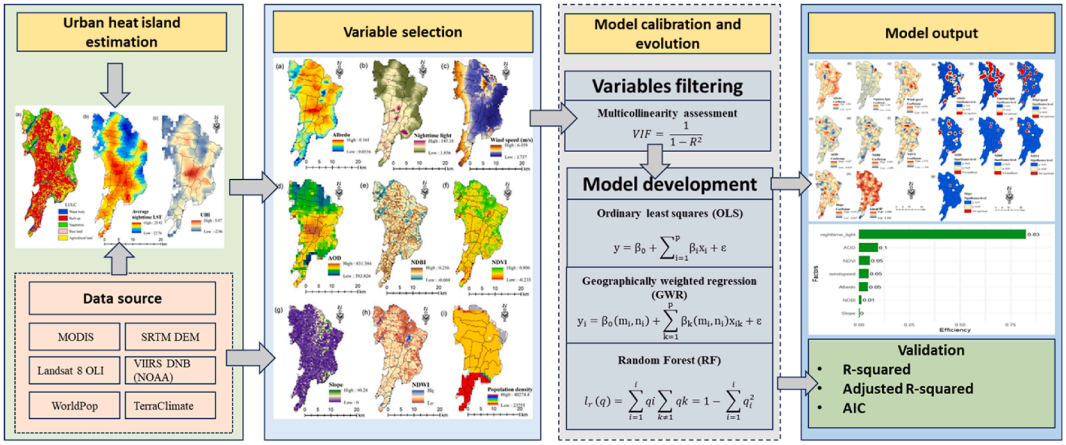


Fig. 2. Methodological flowchart used in this study.

Table 1
Dataset details.

Data	Dataset Name	Data Source Link	Date of Acquisition	Spatial Resolution
Nighttime LST	MOD11A2 (MODIS)	https://MODIS_061_MOD11A2	2024	1 km
Albedo	MCD43A3 (MODIS)	https://MODIS_061_MCD43A3	2024	500 m
LULC	Landsat 8 OLI	https://LANDSAT_LC08_C02_T1_L2	2024	30 m
Nighttime Light	VIIRS DNB (NOAA)	https://NOAA_VIIRS_DNB_MONTHLY_V1_VCMSLCFG	2024	500 m
NDVI	Landsat 8 OLI	https://LANDSAT_LC08_C02_T1_L2	2024	30 m
NDBI	Landsat 8 OLI	https://LANDSAT_LC08_C02_T1_L2	2024	30 m
NDWI	Landsat 8 OLI	https://LANDSAT_LC08_C02_T1_L2	2024	30 m
AOD	MOD04_L2 (MODIS)	https://MODIS_061_MOD04_L2	2024	10 km
PD	WorldPop	https://www.worldpop.org/	2020	1000 m
Wind Speed	TerraClimate	https://IDAHO_EPSCOR_TERRACLIMATE	2024	~4 km
Slope	SRTM DEM	https://CGIAR_SRTM90_V4	2024	30 m

relationships and localised effects of predictor variables on nighttime LST. Additionally, the Random Forest ML algorithm was implemented in R Studio to assess the relative importance of each factor and enhance the predictive modelling of urban nighttime heat intensity.

3.3. Assessment of LST and UHII

To quantify UHII, the nighttime LST data were derived from the MODIS MOD11A2 product at 1 km resolution for the year 2024. The nighttime UHII was calculated by comparing it with the average LST of the urban area and the surrounding rural area (Cheval et al., 2022; Karthikeya et al., 2024; Wong and Nichol, 2013). Mean LST values were extracted for both zones using the zonal statistics tool in ArcGIS, where LST urban represents the mean nighttime LST of urban zones, and LST rural is the mean nighttime LST of non-urban zones. The UHII was computed using the following formula:

$$\mathrm{UHII} = \mathrm{LST\ urban} - \mathrm{LST\ rural} \tag{1}$$

3.4. Identifying and exploring influential factors

3.4.1. Filtering the variables

Firstly, nine independent variables were selected to examine their influence on nighttime heat mitigation across the MMR. These variables were NDVI, NDBI, albedo, wind speed, slope, AOD, nighttime light intensity, NDWI, and population density. To check the multicollinearity among these variables, the Variance Inflation Factor (VIF) analysis was conducted. All the raster images were converted into raster to point, with extracted values first exported into Microsoft Excel and then imported into R Studio for statistical assessment. The VIF calculation was executed using the "car" package in R Studio, precisely the vif () function. A VIF threshold of 5.0 was accepted in this study as the cutoff to detect problematic multicollinearity. The VIF results revealed that NDWI and population density displayed VIF values surpassing the threshold level, indicating a strong linear relationship with other predictor variables. Consequently, these two variables were excluded from further analysis, and the remaining seven variables were retained for subsequent modelling and interpretation.

3.4.2. Model development and evaluation

(a) OLS model

The OLS model is a statistical technique used to estimate the relationship between one or more independent variables and a dependent variable. It functions by reducing the total squared discrepancies between the observed and anticipated values of the dependent variable (Jiao et al., 2021; Koh et al., 2020; Shifti et al., 2020). In this study, the outcome variable is nighttime LST, while the descriptive variables are albedo, wind speed, AOD, Nighttime light, NDVI, NDBI, and slope. This OLS model is driven by the following equation:

$$y = \beta_0 + \sum_{i=1}^{p} \beta_i w_i + \epsilon \tag{2}$$

Where, y indicates the outcome variable, β_0 is a constant, w_i is the ith descriptive variable, β_i is the ith regression coefficient, ε is the error term, and p is the number of explanatory variables.

(b) GWR model

The GWR is one of the important statistical methods for helping to identify how the exposure variables affect the outcome variable at the local or regional level (Brunsdont et al., 1998; Fotheringham et al., 2002; Propastin, 2012). Unlike standard regression models, which assume a constant link between variables across space, GWR allows the relationship to vary geographically. Therefore, weight is

assigned to each observation in GWR according to how closely it fits other observations. These weights are used to evaluate local regression coefficients, resulting in a unique regression equation for each location in the study region (Shao et al., 2020). This model offers a feasible approach to examining how location affects the relationship between variables. Geographically correlated mistakes or changeable associations that vary by region are examples of data exhibiting spatial heterogeneity, in which case GWR is particularly useful (Sun et al., 2021). The GWR is calculated using Equation (3):

$$y_i = \beta_0(m_i, n_i) + \sum_{k=1}^{p} \beta_k(m_i, n_i) x_{ik} + \epsilon \tag{3}$$

where y_i represents the outcome variable for location i, m_i and n_i are the coordinates of location i, $\beta_0(m_i, n_i)$ is the constant at location i, $\beta_k(m_i, n_i)$ is the local parameter estimate for an explanatory variable x_{ik} at location i, and ε is the error term.

(c) RF model

The RF approach is a type of ensemble learning that combines multiple simple decision trees. It is commonly employed for both regression and classification tasks (Dutta et al., 2021). It was initially a random-subspace-based method for random-decision forests (Ho, 1998). Through a trial-and-error approach, the number of trees and variables considered at each split was adjusted to improve model performance. By means of measurements such as the Gini index and the mean decline in accuracy, the RF model also offers further insights, enabling the most important factors influencing urban heat intensity in this area to be identified. The RF model was applied in this study with the "Random Forest" programme. The training phase began with a comprehensive dataset modelling exercise. To assess the generalisability of the model, a 10-fold cross-validation technique was set up using the "trainControl" function from the "caret" R package. Over the "mtry" hyperparameter, a grid search was run to optimise the model.

$$I_r(q) = \sum_{i=1}^{i} qi \sum_{k \neq 1} qk = 1 - \sum_{i=1}^{i} q_i^2 \tag{4}$$

4. Results

4.1. Thermal environment of the MMR

Fig. 3 illustrates the LULC, nighttime LST, and UHII over the MMR, with blue designating water bodies. The LULC classification shows the dominant red patches signifying built-up zones while green, yellow, and pink indicate vegetation, agricultural land, and bare ground, respectively (Zhang et al., 2025). The significant concentration of developed areas in the southern and central parts points to notable urbanisation. At temperatures fluctuating from 22.76 °C to 25.92 °C, the average nightly LST shows warm areas (yellow and orange) that align strongly with built-up areas, suggesting a strong relationship between urban growth and high evening temperatures. On the other hand, colder zones match places with vegetation and water sources. From −2.96 to +5.07, UHII highlights red areas with stronger urban-induced temperature effects. These hotspots match the built-up zones, verifying that urbanisation increases nighttime heat retention. The maps emphasise the significance of sustainable urban design and improved green space infrastructure in the MMR to help reduce thermal stress. The maps show that urban land cover greatly influences nighttime warming and worsens the UHI effect.

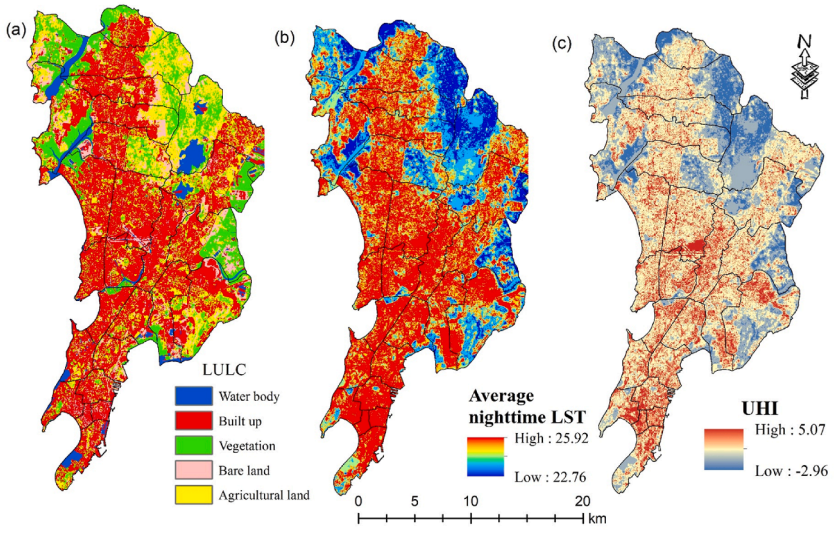


Fig. 3. Spatial distribution of (a) LULC, (b) nighttime LST, and (c) UHI across the MMR.

4.2. Temporal variation of nighttime LST

Fig. 4 illustrates the temporal variation of LST in urban areas of the MMR from 2000 to April 2025. The top graph shows daytime LST, which fluctuates significantly between approximately 25 °C and 42 °C. Peaks correspond to summer months, while dips are associated with monsoon periods or outliers, such as extreme drops seen in 2009, 2016, and 2017. Overall, there is a slightly increasing trend, indicating urban heat accumulation over time. The bottom graph displays nighttime LST, which ranges from about 20 °C to 28 °C, indicating a steadier and less variable pattern compared with daytime LST. However, there is an obvious increasing trend that points to ongoing nighttime warming, a hallmark of the UHI impact. Consistent increased nighttime LST values across the years hint at increasing thermal retention in built-up regions, driven by urbanisation, less vegetation, and infrastructure-induced heat storage.

Fig. 5 illustrates the temporal variation of LST over vegetated areas in the MMR from 2000 to April 2025. The top panel shows the daytime LST, which fluctuates seasonally between 25 °C and 40 °C, with prominent annual peaks during the pre-monsoon period. Over the 25-year period, the general trend remains constant despite seasonal cycles, suggesting that vegetated zones preserve a continuous thermal profile. According to the second graph in Fig. 5, the nighttime LST varies from 18 °C to 25 °C, indicating that nocturnal LST in vegetated areas exhibits little long-term rise, implying that vegetation reduces heat retention. Sometimes, significant declines appear as anomalies, which may be linked to weather phenomena. Vegetated areas exhibit less heat accumulation and better temperature control than metropolitan trends (Fig. 5). These findings underscore the vital role of vegetation in UHI mitigation techniques and support its cooling function.

Fig. 6 illustrates the temporal changes in LST over water bodies in the MMR from 2000 to April 2025. The first graph presents daytime LST, fluctuating between 27 °C and 35 °C, with seasonal peaks. Although some extreme outliers (notably around 2009 and 2016) show abrupt temperature drops, the overall trend is relatively stable, indicating water bodies' capacity to buffer daytime heat. The second graph depicts nighttime LST, ranging from approximately 20 °C to 27 °C. Compared to urban or vegetated areas, water bodies show smoother temperature curves and moderate night temperatures with little long-term increase. The absence of a sharp rising trend suggests that water surfaces retain thermal stability, making them key regulators in the urban thermal landscape. This consistency further supports the cooling function of water bodies in urban areas by lowering the daytime heat and overnight retention that usually causes UHI impacts.

4.3. Exploring the influencing factors of urban heat intensity in Mumbai

Fig. 7 depicts the spatial distribution of nine anthropogenic and environmental variables that together affect the urban thermal climate of the MMR. Regions of low albedo (Fig. 7a) in the urban core receive more solar radiation and experience greater intensification of surface heat, whereas high-albedo coastal and open areas reflect more energy and stay cooler. Increased nighttime light intensity (Fig. 7b) in heavily urbanised regions indicates more human activity and energy use, leading to nighttime warmth. Lower wind speed (Fig. 7c) in the city centre restricts heat dispersal, while higher coastal winds aid in cooling. Increased AOD (Fig. 7d) over industrial and congested traffic areas retains outgoing radiation, further increasing surface temperatures. The NDBI spatial pattern (Fig. 7e) is consistent with areas of higher LST, while regions with greater vegetative cover, as indicated by the NDVI (Fig. 7f), exhibit a cooling effect due to evapotranspiration. Land surfaces with lower elevation and minimal slope (Fig. 7g) retain heat effectively due to limited air movement. In contrast, elevated areas allow for ventilation and cooling. The NDWI (Fig. 7h) demonstrates the cooling effect of water bodies and moisture-rich areas, which can moderate the surrounding temperature. Finally, densely populated areas (Fig. 7i) in the southern and central MMR are associated with significant anthropogenic heat emissions, as well as minimal vegetation and dense

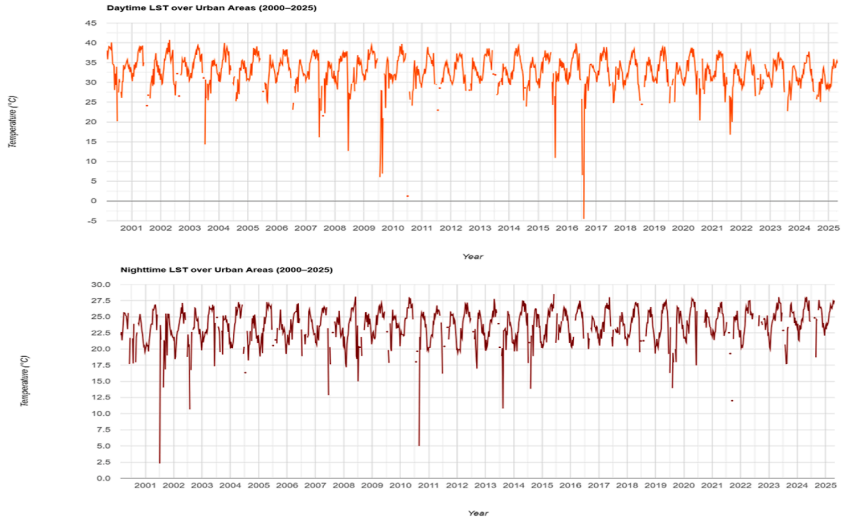


Fig. 4. Temporal changes in LST over urban built-up areas from 2000 to April 2025 in the MMR.

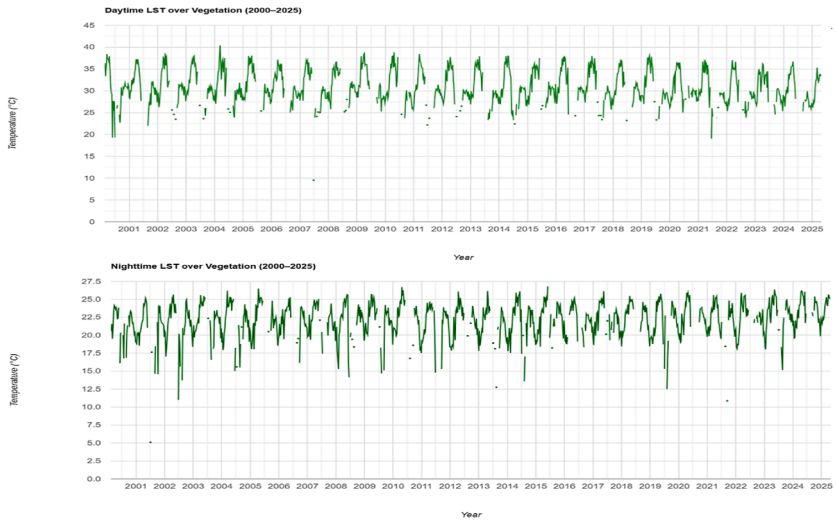


Fig. 5. Temporal changes in LST over vegetation-covered areas from 2000 to April 2025 in the MMR.

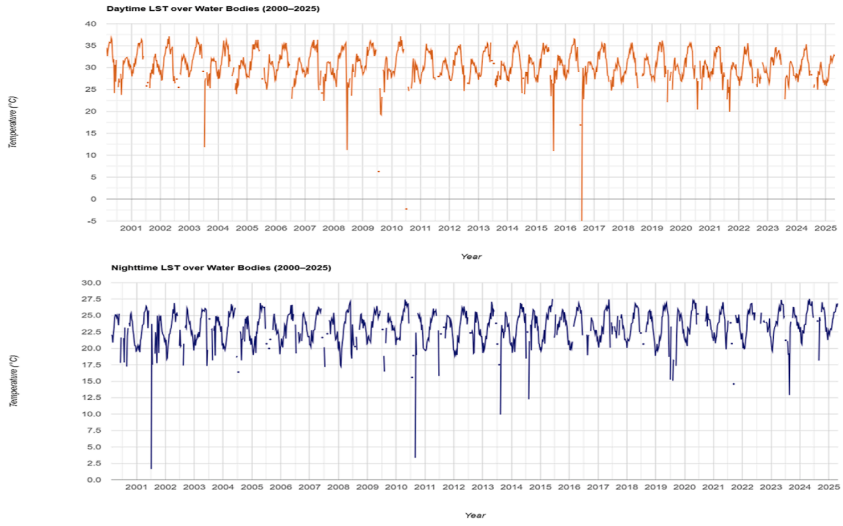


Fig. 6. Temporal changes in LST over water bodies from 2000 to April 2025 in the MMR.

building infrastructure, which altogether enhance the UHI effect. Thus, these spatial patterns illustrate that built-up, polluted, and poorly ventilated areas experience more thermal stress than vegetated and water-abundant areas, while well-aerated areas serve as natural mitigators of urban heat accumulation.

4.4. Assessment of OLS

The OLS regression results provide insights into the effect of various environmental and urban factors on nighttime LST across the MMR (Table 2). The model intercept is statistically significant ($p < 0.0001$), with a value of 23.475, representing the baseline nighttime LST when all predictors are zero. Among the independent variables, nighttime light exhibits the strongest positive effect (coefficient = 0.0289, $t = 45.64$, $p < 0.0001$), confirming its dominant role in elevating nighttime temperatures in urbanised zones. Albedo (coefficient = 4.001, $p < 0.0001$) also contributes positively, suggesting that reflective surfaces in urban areas may trap heat rather than mitigate it. Conversely, NDVI exhibits a strong negative relationship (coefficient = −0.699, $p < 0.0001$), indicating that vegetation significantly contributes to a reduction in heat intensity. Wind speed and slope are also negatively correlated with LST, suggesting that natural topography, air movement, and built-up characteristics influence thermal variations. The positive but modest effect of AOD (coefficient = 0.0002, $p = 0.0314$) highlights the heat-trapping potential of aerosols. All variables exhibit low multi-collinearity (VIF <2), confirming the model's reliability. Overall, the OLS model effectively identifies both the anthropogenic and natural factors influencing UHI.

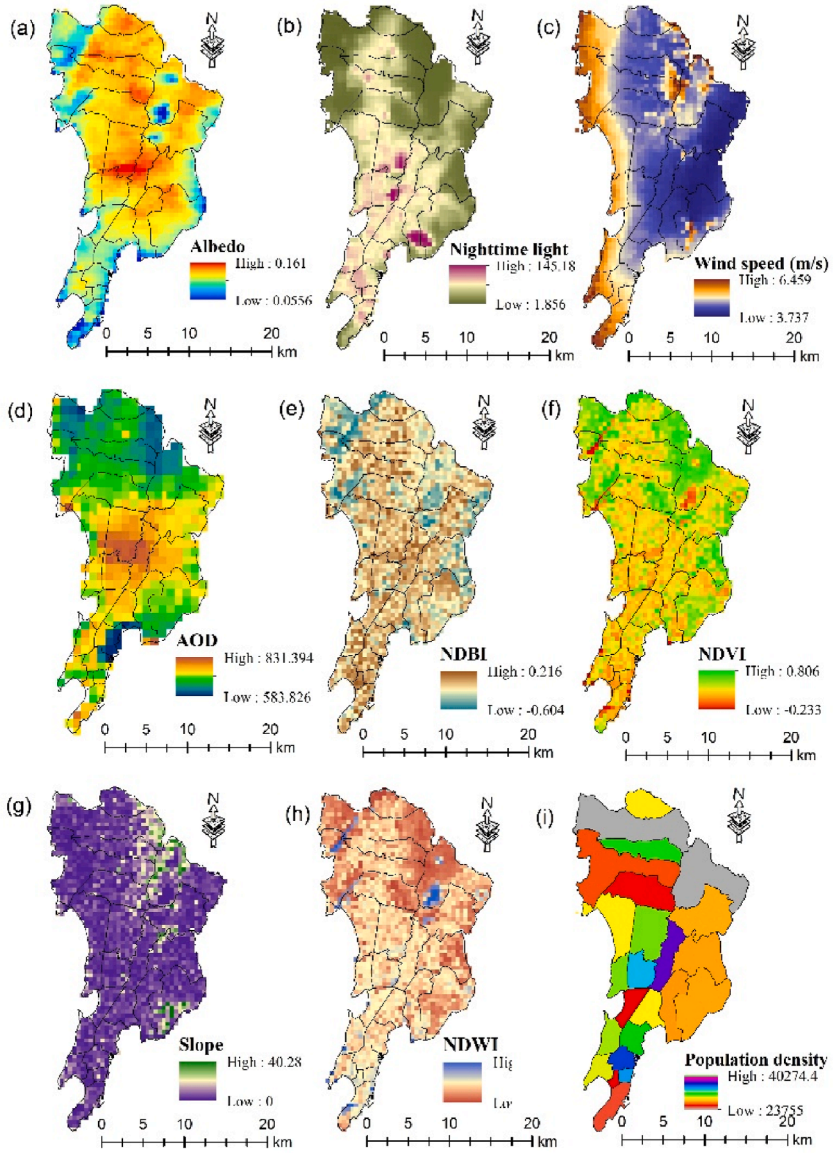


Fig. 7. Spatial distribution of the variables influencing nighttime LST in the MMR.

Table 2
Regression coefficients of the OLS model.

Variable	Coefficient	Std.Error	t-statistic	Probability	VIF
Intercept	23.475	0.1322	177.5317	0.0000	–
Albedo	4.001	0.9398	4.2567	0.0000	1.283
Wind speed	−0.037	0.0141	−2.6238	0.0087	1.034
AOD	0.0002	0.00009	2.1532	0.0314	1.123
Nighttime light	0.0289	0.0006	45.6372	0.0000	1.578
NDVI	−0.699	0.0749	−9.3218	0.0000	1.643
NDBI	0.2567	0.1032	2.4866	0.0130	1.521
Slope	−0.0091	0.0022	−4.0494	0.0000	1.105

Table 3 displays the Pearson correlation coefficients among the same set of variables, offering insight into their pairwise linear relationships with LST. Consistent with the OLS findings, nighttime light exhibits a robust, positive correlation with LST ($r = 0.826$, $p < 0.01$), reinforcing its dominant role in urban heating. AOD and NDBI also exhibit moderate positive correlations with LST

(r = 0.478 and 0.340, respectively, both $p < 0.01$), while NDVI, slope, and wind speed demonstrate weak to moderate negative correlations (r = −0.052 to −0.256, all $p < 0.05$), affirming their role in reducing LST. These correlation values support the regression outputs and emphasise the complex interplay between built-up intensity, topography, vegetation, and atmospheric components in shaping nighttime heat distribution.

4.5. Assessment of GWR

Fig. 8 presents the spatial distribution of GWR coefficients and local R^2 values for the variables influencing nighttime LST across the MMR. Each sub-map (a–g) shows how the effect of a variable varies spatially, highlighting the non-stationarity of influencing factors.

Albedo has a positive effect in the south and middle MMR, which means locally reflective surfaces enhance LST. Nightlight is highly associated with increased LST in north-central regions, validating its potential as an indicator of urban activity and heat emission. Wind speed is negatively associated, especially in the east, implying efficient heat dissipation. AOD presents mixed effects with a relatively stronger warming in the west as a result of pollution-driven heat trapping. NDBI has a positive effect on LST throughout the MMR's central region, highlighting the effect of compact built-up surfaces, while NDVI has consistent negative effects, underscoring vegetation's cooling effect. Slope plays a weak role in cooling, with spatial variation. The local R^2 measures up to 0.93, indicating strong spatial heterogeneity among LST drivers.

Fig. 9 shows the spatial importance of variables that drive nighttime LST according to the GWR model. Nightlight is significantly important in the northern and northeastern MMR, indicating intensive urbanisation. Wind speed is locally and irregularly significant, while NDVI and NDBI are still significant in most areas, validating their prevailing cooling and warming effects. Albedo, AOD, and slope also demonstrate considerable importance in urban and transition areas. The high prevalence of highly significant areas supports strong spatial dependence of LST with respect to environmental and anthropogenic variables, providing valuable information for spatially focused UHI mitigation in the MMR.

4.6. Assessment of RF

Fig. 10 highlights the variable importance in the RF model. Nighttime light is the dominant predictor, with an efficiency score of 0.83, far surpassing all other factors. This confirms that urban activity and infrastructure, indicated by nighttime illumination, are the primary drivers of elevated nighttime LST, while AOD (0.10), NDVI, wind speed, and albedo (each 0.05) also contribute moderately. In contrast, NDBI (0.01) and slope (0.00) have minimal impact on the RF model despite showing significance in spatial or linear models.

4.7. Model performance assessment

Table 4 compares the model performance. The GWR model performs best, exhibiting the highest R-squared (0.9808) and Adjusted R-squared (0.9539), indicating it captures local spatial variations more effectively. The RF model also performs well (R-squared = 0.7985), better than OLS (R-squared = 0.7067), but lacks AIC for comparison. The AIC score, available only for parametric models, shows that the GWR (666.80) fits the data much better than OLS (2285.99), reinforcing the advantage of spatially adaptive modelling.

5. Discussion

This study provides key insights into the growing problem of the nighttime UHI effect and temperature differences in the MMR. The findings show that fast urban growth, more paved surfaces, and higher artificial light affect the city's nighttime heat. These results match earlier studies reporting that urban growth leads to more heat being retained at night by changing how surfaces hold energy (Aslani et al., 2025; Gao et al., 2020; Hasan et al., 2025; Maity et al., 2025; Oke, 1982). In addition, similar patterns have also been observed in other major tropical cities, including Bangkok (Karthikeya et al., 2024; Varnakovida & Ko, 2023), Jakarta (Siswanto et al., 2023), and Delhi (Kumari et al., 2021), where rapid and uncontrolled urban expansion has contributed to elevated levels of nighttime surface temperatures. In this study, vegetation cover and water bodies play a significant role in maintaining a cooling environment

Table 3
Pearson correlation coefficients for the variables.

Factors	LST	Albedo	Windspeed	AOD	Nighttime_Light	NDVI	NDBI	Slope
LST	1	0.237[a]	−0.243[a]	0.478[a]	0.826[a]	−0.052[b]	0.340[a]	−0.256[a]
Albedo	0.237[a]	1	−0.430[a]	0.226[a]	0.253[a]	−0.070[a]	0.372[a]	0.081[a]
Windspeed	−0.243[a]	−0.430[a]	1	−0.294[a]	−0.179[a]	0.044	−0.097[a]	0.117[a]
AOD	0.478[a]	0.226[a]	−0.294[a]	1	0.419[a]	0.036	0.146[a]	−0.209[a]
Nighttime light	0.826[a]	0.253[a]	−0.179[a]	0.419[a]	1	−0.037	0.430[a]	−0.223[a]
NDVI	−0.052[b]	−0.070[a]	0.044	0.036	−0.037	1	−0.085[a]	0.034
NDBI	0.340[a]	0.372[a]	−0.097[a]	0.146[a]	0.430[a]	−0.085[a]	1	0.032
Slope	−0.256[a]	0.081[a]	0.117[a]	−0.209[a]	−0.223[a]	0.034	0.032	1

[a] Significance level at 0.01.
[b] Significance level at 0.05.

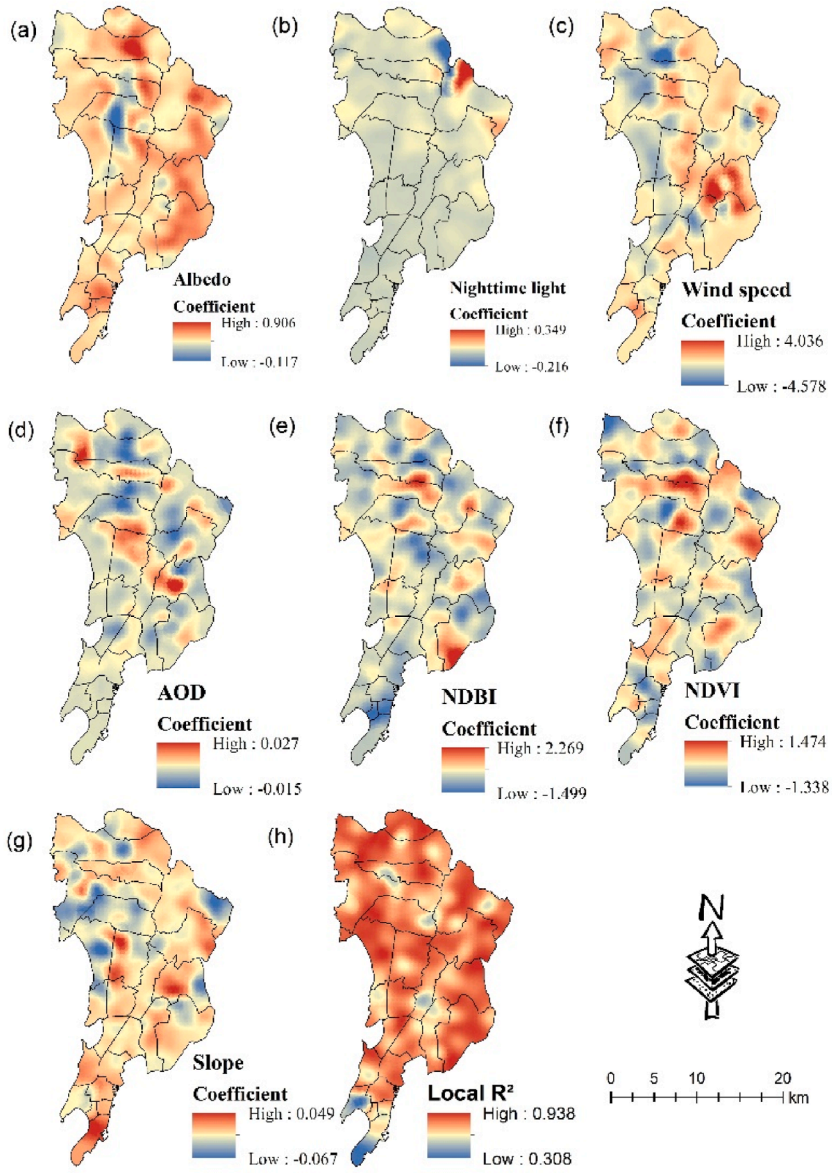


Fig. 8. Spatial distribution of coefficients and local R^2 values for the variables influencing nighttime LST in the MMR, derived using the GWR model.

through evapotranspiration and increased surface permeability. Previous studies have also reported similar findings, indicating that vegetation cover and water bodies display a negative association with surface LST, while their surrounding areas exhibit lower temperatures in urban areas (Bekele et al., 2022; Chowdhury et al., 2025; Hasan et al., 2025; Fan et al., 2025). A study conducted by Uddin et al. (2022) in the Dhaka metropolitan region highlights that in the city centre, vegetation cover is very high, which reduces the nighttime LST by up to 2 °C. Another study from Shanghai (Wang et al., 2017) and Singapore (Mughal et al., 2020) found that the spatial endurance of vegetation significantly enhances nighttime cooling efficiency. This study also reports similar results, indicating a negative correlation between NDVI and LST. The GWR analysis reveals that the exposure variables influence nighttime LST differently across the MMR due to spatial heterogeneity in the impact of these variables. In this study, the results highlight that the GWR model has the highest R^2 values, indicating that the interaction between the exposure variable and outcome variable is best fitted on a local scale rather than a global model, such as OLS and RF. The results of the study align with studies conducted in Seoul by Jang et al. (2020) and Guangzhou by Liu et al. (2024), indicating that spatially adaptable models can elucidate urban temperature variations across different land cover types. The high level of anthropogenic factors, particularly nighttime illumination and urban density, underscores the significance of human-induced radiative forcing in shaping the thermal structures of metropolitan areas. Similar relationships between nighttime light reflections and surface LST have been identified in Beijing (Li et al., 2022) and Chennai (Bera et al., 2022), where artificial lighting serves as both an indicator of economic activity and a direct measure of surface temperature. Additionally, the

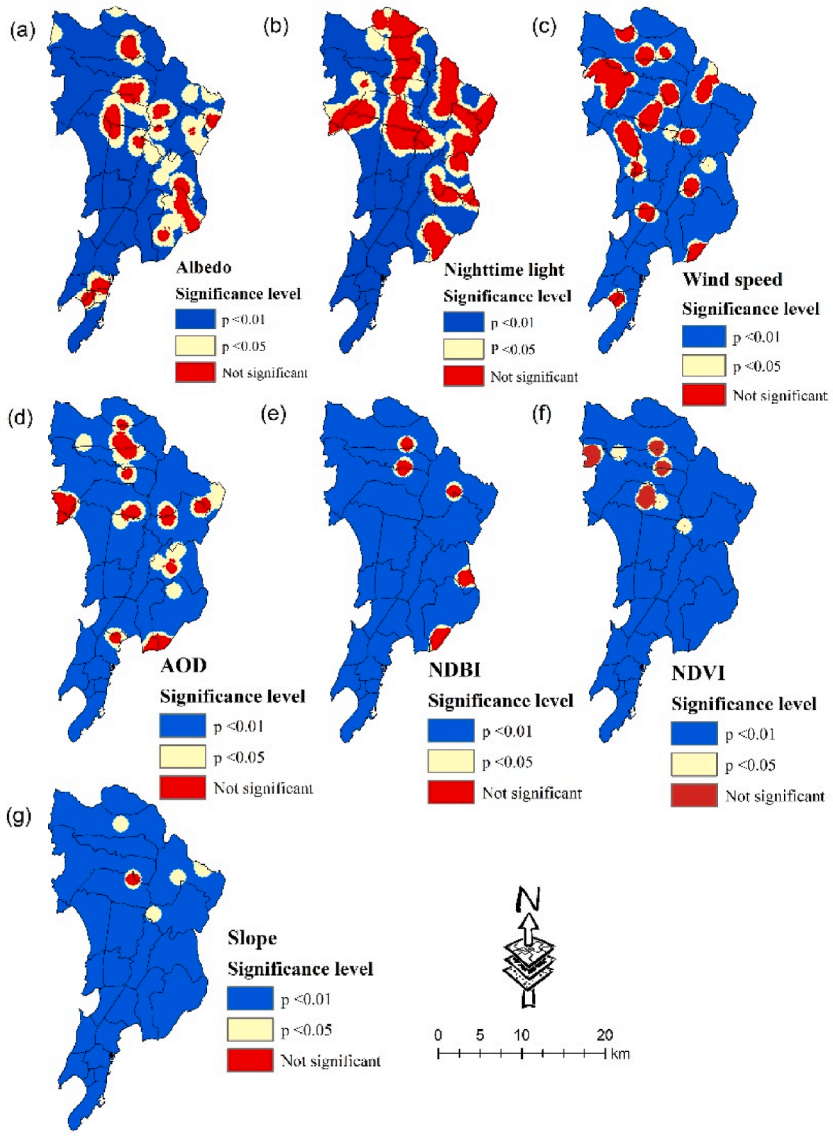


Fig. 9. Spatial distribution of the significance level for variables influencing nighttime LST in the MMR, derived using the GWR model.

contribution of aerosols (AOD) is directly correlated with LST, consistent with similar results reported for Delhi and Kolkata, where particulate air pollution traps outgoing radiation and contributes to enhanced nighttime heat maintenance (Dahutia et al., 2018; Dutta & Chatterjee, 2022; Manna & Sarkar, 2025; Srivastava et al., 2012). Additionally, these comparisons reinforce the acknowledgment of anthropogenic factors in intensifying the UHI effect in rapidly urbanising Asian cities. From a planning perspective, this study underscores the urgent need for integrated mitigation strategies that incorporate both natural and anthropogenic solutions. The "Green Delhi Mission" and the "Singapore Cooling Plan" demonstrate that integrating additional parks and green spaces, street trees, and rooftop vegetation can help mitigate heat stress (Pramanik & Punia, 2019; Li et al., 2024; Ramsay et al., 2025). Alongside this, the advancement of reflective and high-albedo materials under Mumbai's Development Plan (DP) 2034 and Mumbai Climate Action Plan (MCAP) could mitigate heat absorption and activity, thus reducing heat capacity (Mumbai Climate Action Plan, nd). The mitigation of artificial lighting through smart illumination systems, as observed in the cases of Seoul and Tokyo (Jang et al., 2020), is another potential approach for reducing heat buildup during nighttime while maintaining safety in the urban realm.

Finally, this research contributes to the understanding of the spatial variation in nighttime urban climate by employing spatial modelling and machine learning to demonstrate the complexities of nighttime LST. The results indicate that anthropogenic nighttime light and building density are the primary factors causing heat accumulation, whereas vegetation cover and wind direction are significant factors that mitigate thermal stress. The results substantiate the statements made by Pramanik et al. (2025) and Moharir et al. (2025) in that evidence-based interventions are crucial for enhancing urban resilience in the face of escalating climate change. This study contributes to the growing body of urban climate literature by establishing a transportable analytical framework that

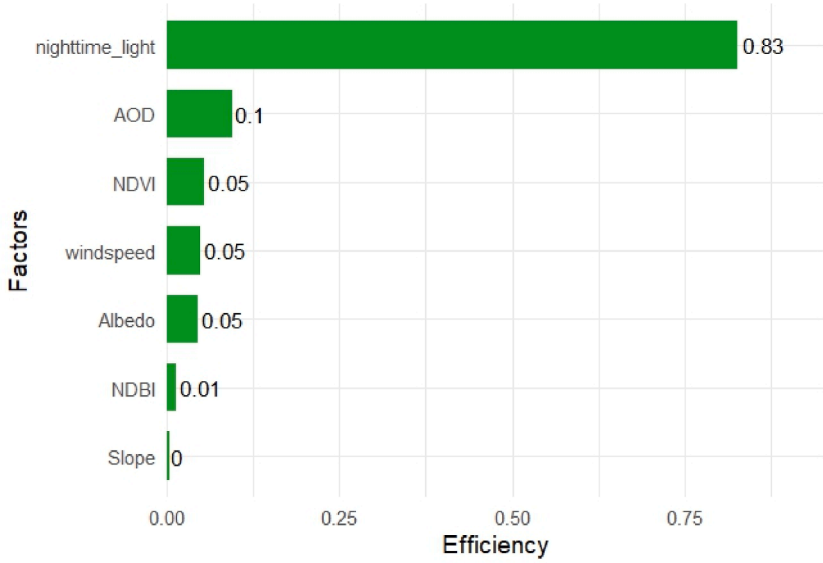


Fig. 10. Results analysis using the RF Model.

Table 4
Comparison of the goodness of fit measures for the OLS, GWR, and RF models.

Criterion	OLS	GWR	RF
R-squared	0.7067	0.9808	0.798538
Adjusted R-squared	0.7055	0.9539	0.79452
AIC	2285.9949	666.8034	–

incorporates both methodological and empirical insights. It also presents a real-world model for mitigating the LST and UHI effects in rapidly growing cities in India and the Global South.

6. Conclusion and policy recommendation

This study presents an in-depth examination of the increasing nighttime LST and UHI intensities in the MMR over the past two decades. The study reveals that the southern and central parts of the city, particularly in densely populated areas, places with limited vegetation, and locations with excessive artificial light, experience the highest nighttime temperatures. These places have temperature gradients of +5.07 °C, highlighting the intensity of nighttime urban warming. This illustrates the significant contribution of unplanned urban growth to thermal stress. The results align with the idea that human-made factors, such as nighttime lights and built-up areas, are the primary causes of urban warming, while natural factors, including plants, water bodies, and wind circulation, help to cool things down. The use of spatially adaptable models, such as the GWR, demonstrates the distinct variations in these characteristics across different locations, highlighting the importance of targeted adjustments in specific areas to enhance the resilience of cities to heat.

This study emphasises a comprehensive strategy to mitigate the escalating UHI effect, incorporating nature-based remedies, technological investments, and evidence-based planning. New parks, green corridors, street trees, rooftop gardens, or vertical vegetation atop buildings in densely populated metropolitan areas should be prioritised for urban greening. The strategy can also focus on scaling reflective materials, cool roofs, and high-albedo pavements, part of the Development Plan (DP) 2034 and the MCAP. These materials help to keep buildings from absorbing too much heat. Smart lighting systems can also control artificial light, and ensuring sufficient ventilation corridors helps to improve airflow in cities and mitigate the impact of the next heat wave at night. Integrating real-time GIS and remote sensing data-driven observation into urban government could enable ongoing monitoring and flexible decision-making. To make cities more climate-resilient and give everyone equal access to cooling or communal green spaces or cooling centres, people who live in vulnerable informal settlements should be included in the call for intentional community-level cooling intervention to help reduce the UHI effect. These initiatives will make metropolitan Mumbai more resilient to climate change and serve as a model for controlling heat in a manner that benefits rapidly growing cities in India and the Global South.

Mumbai policy mechanisms should eventually include ongoing monitoring and decision-making based on data. As part of the Smart City Mission, the city uses GIS technology for urban planning and satellite-based remote sensing technologies. This has made it possible to keep an eye on UHI hotspots in real time. Continuous use of these technologies, along with collecting environmental data, can help with adaptive planning and targeted measures in places most affected by the UHI effect. It is extremely important to make sure that

lowering the UHI effect will help low-income communities and is done in a way that includes everyone, as this is a question of social justice. Informal settlements most afflicted by heat slowly become used to these conditions because they are usually in the least green areas without a good infrastructure. The policies at the top of the list should be those that make it possible for everyone to use cooling measures, such as creating community cooling centres and green roofs. The MCAP already recognises the importance of inclusive participation in climate action. Consequently, the policies that come next need to ensure that UHI reduction programmes address the requirements of the most vulnerable people. These suggestions are not only important for Mumbai, but also demonstrate a deep awareness of other rapidly growing cities in India and the Global South. This strengthens their position in the global fight against climate change and in building cities that will last.

CRediT authorship contribution statement

Harekrishna Manna: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Malay Pramanik:** Writing – review & editing, Visualization, Supervision, Investigation, Formal analysis, Conceptualization. **Rizwan Ahamed:** Writing – original draft, Visualization, Formal analysis, Conceptualization. **Sanjit Sarkar:** Writing – review & editing, Supervision, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that there are no known competing financial interests or personal ties that could have influenced the work described in this paper.

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